AI Game Mathematics Assistant Business & Technical Documentation

Executive Summary

AI Game Mathematics Assistant is a SaaS platform for iGaming providers to design, validate, simulate, analyze, and explain slot game mathematics before certification and release.

Problem Statement

Game studios rely heavily on spreadsheets, manual simulations, fragmented tools, and expert mathematicians. The platform centralizes validation, simulation, RTP analysis, and AI-powered explainability.

Target Customers

Small/Medium slot studios, independent math consultants, game designers, certification preparation teams, iGaming startups.

Value Proposition

Reduce development time, improve mathematical accuracy, provide explainable insights, lower dependency on specialist resources.

Pricing

Starter $299/month (10 simulations/day); Professional $999/month (unlimited simulations, AI insights); Enterprise $2,999+/month (multi-user, SSO, on-premise). Consulting services: $5,000–$20,000/project.

Core Features

Game Definition DSL; Validation Engine; Runtime Engine; Simulation Engine; Mathematical Engine; AI Explainability Assistant; Golden Fixtures; Reporting.

Tech Stack

Backend: Python 3.13, FastAPI, Pydantic v2, PostgreSQL, Redis. Frontend: React, TypeScript, Tailwind. Infrastructure: Docker, GitHub Actions. AI: OpenAI API.

Architecture

Modular monolith architecture with bounded contexts: Validation, Runtime, Simulation, Mathematics, Plugins, API, AI Assistant.

Third-party Tools

OpenAI API, PostgreSQL, Redis, Docker, uv, Pytest, Ruff, MyPy, Black, Structlog.

Database Schema

Organizations, Users, Projects, GameVersions, SimulationRuns, SimulationResults, ValidationReports, GoldenFixtures, AIExplanations.

Domain Models

GameVersion, StateMachine, State, Transition, PluginDefinition, SimulationProfile, ValidationReport, SpinChainResult.

Validation Engine

Schema validation, graph validation, plugin validation, exposure validation, runtime limits validation.

Runtime Core

Deterministic FSM execution with push/pop state machines, plugin host, RNG service, exposure enforcement, event generation.

Simulation Engine

ProcessPoolExecutor-based parallel simulations with deterministic seed management and shard aggregation.

Mathematical Engine

RTP, hit frequency, volatility, exposure calculations using integer arithmetic and NumPy statistical functions.

AI Assistant

Explains RTP changes, compares simulations, identifies anomalies, assists game designers.

Security

Role-based access control, audit logging, encrypted secrets, isolated simulations.

Roadmap

MVP → Advanced Mechanics → Optimization AI → Marketplace & Enterprise Features.

High-Level Architecture

Frontend	React + TypeScript
API Layer	FastAPI
Validation Engine	JSON Schema + Graph Validators
Runtime Core	Deterministic FSM Engine
Simulation Engine	Multiprocessing Workers
Mathematical Engine	RTP & Volatility Analytics
AI Assistant	OpenAI-powered Explainability
Storage	PostgreSQL + Redis

Database Entities

Entity	Purpose
Organization	Customer account
User	Platform users
Project	Game development workspace
GameVersion	Immutable game snapshot
SimulationRun	Simulation metadata
SimulationResult	Aggregated outputs
ValidationReport	Validation findings
GoldenFixture	Regression testing
AIExplanation	Generated insights